

CV of BAO Sheng

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Research Interest

Embodied Intelligence; Transportation; Robotics and Autonomous Systems; Multi-source Data Fusion; Digital Twin

Academic Position

2024.07-present: **Postdoctoral Fellow & Part-time Lecturer** (from 2025.09) The University of Hong Kong (HKU)

2022.12-2024.07: **Postdoctoral Fellow** The Hong Kong Polytechnic University (PolyU)

Academic Qualification

2019.09-2022.11 **Ph.D.** The Hong Kong Polytechnic University
Major: Urban Informatics and Smart City

2015.09-2018.04 **M.Eng.** Nanjing University of Aeronautics and Astronautics
Major: Control Theory and Control Engineering

2011.09-2015.06 **B.Eng.** Nanjing University of Aeronautics and Astronautics
(Top 50 universities in China)
Major: Automation Engineering

Publications (* indicates corresponding author, 1 indicates co-first author)

1. **S. Bao**, W. Pan, Z. Zheng, and B.P.Y Loo, "Ultra-long and Heavy Vehicles on Roads: Quantifying Disruptions on Road Traffic Using Fully Vision-based Methods." *Transportation Research Part C: Emerging Technologies*, Elsevier. (Submitted)
2. Z. Huang, J. Zhou, B. Zhou, H. Xiao, **S. Bao***, and B.P.Y Loo, "A Socio-technical Transition of the Low-Altitude Economy: Evidence and Governance Implications from Chinese Cities," *Sustainable Futures*, Elsevier. (Under Review)
3. **S. Bao**, J. Zhou, W. Pan, Z. Zheng, and B.P.Y Loo, "Combining emerging technologies with transport impact analysis: A video-based assessment framework for dynamic interaction with long and heavy vehicles." *IEEE Transactions on Intelligent Transportation Systems*, IEEE. (Under Review)
4. S. Lu, **S. Bao**¹, Y. Wei, S. Zhang, D. Yang, and W. Shi, "Multimodal sensor dataset from vehicle-mounted mobile mapping system for comprehensive urban scenes," *Scientific Data*, vol. 12, Art no. 1411, 2025, Nature Portfolio. (MULTIDISCIPLINARY SCIENCES: 16/134 Q1; IF: 5.8)
5. **S. Bao**, W. Shi, D. Yang, H. Xiang, and Y. Yu, "Global Principal Planes Aided Mobile Mapping Method in Artificial Environments," *Advanced Engineering Informatics*, vol. 61, Art no. 102472, 2024, Elsevier. (ENGINEERING, MULTIDISCIPLINARY: 4/90 Q1; IF: 8.8)
6. H. Xiao, **S. Bao**, J. Ren, Z. Xu, S. Xue, and J. Liu, "Global transboundary synergies and trade-offs among Sustainable Development Goals from an integrated sustainability perspective," *Nature Communications*, vol. 15, no. 1, Art no. 500, 2024, Nature Portfolio.

(MULTIDISCIPLINARY SCIENCES: 6/73 Q1; IF: 16.6) (Highly Cited Papers)

7. **S. Bao**, W. Shi, P. Chen, H. Xiang, and Y. Yu, "A systematic mapping framework for backpack mobile mapping system in common monotonous environments," *Measurement*, vol. 197, Art no. 111243, 2022, Elsevier. (ENGINEERING, MULTIDISCIPLINARY: 16/90 Q1; INSTRUMENTS & INSTRUMENTATION: 9/63 Q1; IF:5.6)

8. **S. Bao**, W. Shi, P. Chen, and H. Xiang, "Sole GNSS Sensor Extrinsic Calibration Methods for Mobile Mapping Systems," *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 11338-11345, 2022, IEEE. (ROBOTICS: 10/30 Q2; IF:5.2)

9. **S. Bao**, W. Shi, W. Fan, P. Chen, M. Nie, and H. Xiang, "A tight coupling mapping method to integrate the ESKF, g2o, and point cloud alignment," *The Journal of Supercomputing*, vol. 78, no. 2, pp. 1903-1922, 2021, Springer. (COMPUTER SCIENCE, HARDWARE & ARCHITECTURE: 22/54 Q2; COMPUTER SCIENCE, THEORY & METHODS: 37/111 Q2; IF:3.3)

10. F. Guan, K. Tang, J. Zhang, **S. Bao***, L. Chen, R. Chen, and Y. Yu, "Autonomous wireless positioning system using crowdsourced Wi-Fi fingerprinting and self-detected FTM stations," *Expert Systems with Applications*, vol. 255, Art no. 124566, 2024, Elsevier. (COMPUTER SCIENCE, ARTIFICIAL INTELLIGENCE: 24/197 Q1; ENGINEERING, ELECTRICAL & ELECTRONIC: 25/352 Q1; IF: 7.5)

11. H. Xiao, **S. Bao**, J. Ren, and Z. Xu, "Transboundary impacts on SDG progress across Chinese cities: A spatial econometric analysis," *Sustainable Cities and Society*, vol. 92, Art no. 104496, 2023, Elsevier. (CONSTRUCTION & BUILDING TECHNOLOGY: 1/68 Q1; ENERGY & FUELS: 13/115 Q1; IF:11.7)

12. H. Xiao, Z. Xu, J. Ren, Y. Zhou, R. Lin, **S. Bao**, L. Zhang, S. Lu, C. K. M. Lee, and J. Liu, "Navigating Chinese cities to achieve sustainable development goals by 2030," *Innovation*, vol. 3, no. 5, Art no. 100288, 2022, Elsevier. (MULTIDISCIPLINARY SCIENCES: 5/133 Q1; IF:33.1)

13. W. Shi, P. Chen, M. Wang, **S. Bao**, H. Xiang, Y. Yu, and D. Yang, "PolyU-BPCoMa: A dataset and benchmark towards mobile colorized mapping using a backpack multisensorial system," *International Journal of Applied Earth Observation and Geoinformation*, vol. 112, Art no. 102962, 2022, Elsevier. (REMOTE SENSING: 5/33 Q1; IF:7.5)

14. H. Xiang, X. Zhu, W. Shi, W. Fan, P. Chen, and **S. Bao**, "DeLightLCD: A Deep and Lightweight Network for Loop Closure Detection in LiDAR SLAM," *IEEE Sensors Journal*, vol. 22, no. 21, pp. 20761-20772, 2022, IEEE. (ENGINEERING, ELECTRICAL & ELECTRONIC: 91/275 Q2; INSTRUMENTS & INSTRUMENTATION: 15/63 Q1; IF:4.3)

15. Y. Yu, W. Shi, R. Chen, L. Chen, **S. Bao**, and P. Chen, "Map-Assisted Seamless Localization Using Crowdsourced Trajectories Data and Bi-LSTM Based Quality Control Criteria," *IEEE Sensors Journal*, vol. 22, no. 16, pp. 16481-16491, 2022, IEEE. (ENGINEERING, ELECTRICAL & ELECTRONIC: 91/275 Q2; INSTRUMENTS & INSTRUMENTATION: 15/63 Q1; IF:4.3)

16. P. Chen, W. Shi, W. Fan, H. Xiang, and **S. Bao**, "RectMatch: A novel scan matching method using the rectangle-flattening representation for mobile LiDAR systems," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 180, pp. 191-208, 2021, Elsevier. (GEOGRAPHY, PHYSICAL: 1/49 Q1; GEOSCIENCES, MULTIDISCIPLINARY: 4/201 Q1; IF:12.7)

17. H. Xiang, W. Shi, W. Fan, P. Chen, **S. Bao**, and M. Nie, "FastLCD: A fast and compact loop closure detection approach using 3D point cloud for indoor mobile mapping," *International*

Journal of Applied Earth Observation and Geoinformation, vol. 102, Art no. 102430, 2021, Elsevier. (REMOTE SENSING: 5/33 Q1; IF:7.5)

18. P. Chen, W. Shi, **S. Bao**, M. Wang, W. Fan, and H. Xiang, “Low-Drift Odometry, Mapping and Ground Segmentation Using a Backpack LiDAR System,” *IEEE Robotics and Automation Letters*, vol. 6, no. 4, pp. 7285-7292, 2021, IEEE. (ROBOTICS: 10/30 Q2; IF:5.2)

Patents

1. **S. Bao**, P. Lv, J. Lai, and S. Liu, “A method of using Kalman filter to identify the parameters of multi-rotor UAV dynamic model” (**Granted**, Chinese Patent No.: ZL 2017 1 0367495.5)
2. P. Lv, **S. Bao**, J. Lai, P. Shi, C. Yuan, and L. Fan, “A multi-information fusion real-time wind speed estimation method for multi-rotor UAV” (**Granted**, Chinese Patent No.: ZL 2017 1 0073238.0)
3. W. Shi, **S. Bao**, W. Fan, and M. Nie, “A tight coupling mapping method to integrate the ESKF, g2o, and point cloud alignment” (**Granted**, Chinese Patent No.: ZL 2020 1 1351405.1)
4. W. Shi, **S. Bao**, and Y. Yu, “A method, system, and storage medium for step detection and pedestrian dead reckoning” (**Granted**, Chinese Patent No.: ZL 2021 1 0241308.5)
5. W. Shi and **S. Bao**, “A method, device, terminal, and storage medium for GNSS sensors calibration” (**Granted**, Chinese Patent No.: ZL 2021 1 0904029.2)
6. W. Shi, **S. Bao**, H. Xiao, and Y. YU, “A LiDAR-based mobile mapping method and device using global principal planes” (Published, Chinese Patent Publication No.: CN 117455763A)
7. W. Shi, **S. Bao**, and Y. YU, “A method for correcting LiDAR odometry using visual tags” (Published, Chinese Patent Publication No.: CN 116381651A)
8. W. Shi, Y. YU, and **S. Bao**, “A fusion positioning method, system, and terminal based on dual IMU and ultrasonic ranging” (Published, Chinese Patent Publication No.: CN 118111425A)
9. W. Shi, Y. YU, and **S. Bao**, “A map-assisted indoor and outdoor seamless positioning method, system, and terminal” (Published, Chinese Patent Publication No.: CN 118149811A)
10. **S. Bao**, B.P.Y. Loo, W. Pan, and Z. Zheng, “A Deep Learning-based Framework for Recovering 3D Motion Information from Visual Data” (Published, Chinese Patent Publication No.: CN 121482081A; Hong Kong Standard Patent Application No.: 42026119013.6).
11. **S. Bao**, B.P.Y. Loo, W. Pan, and Z. Zheng, “A Method for Estimating Sidewalk Widths Using Street-View Images” (Published, Chinese Patent Publication No.: CN 121527731A; Hong Kong Standard Patent Application No.: 42026119414.6).

Academic Service

- ◆ **Invited speaker (session keynote lecturer)** of the 2025 International Conference on Intelligent Traffic and Transportation
- ◆ **Guest editor** of the SI “Cooperative Perception for Modern Transportation” in *Drones* (REMOTE SENSING: 14/62 Q1; IF: 4.4)
- ◆ Reviewer of *AEI*, *IEEE T-ITS*, and *IJDE*, etc.
- ◆ Excellent Youth Report at the 4th (2024) Spatial Information Industry and International Standard Conference
- ◆ Fellow, Joint Laboratory on Future Cities and The Institute of Transport Studies, HKU

Awards

Recipient of **Hubei Provincial Hundred Talents Program (2025)**

Merit Award of 2023 China Postdoctoral Innovation & Entrepreneurship Competition (Intelligent Protective Cane with adaptive position estimation) (**Leader**)

Gold Medal Award of 2026 International Exhibition of Inventions Geneva (Sight2Insights: A Scalable AI Platform on Urban Visual Intelligence) (Ranked 2nd)

Gold Medal Award of 2025 China Postdoctoral Innovation & Entrepreneurship Competition (Xuntu: A High-Precision Real-Time Indoor and Outdoor Positioning System) (Ranked 3rd)

Best Innovation Design Award of 2016 International Aerial Robotics Competition (Asia-Pacific Venue)

Excellent Graduate Student of Nanjing University of Aeronautics and Astronautics (one of 18 awardees university-wide)

Teaching Experience

- ◆ 2025 **Part-time Lecturer** of HKU GEOG 7014 “GIS in Transport Planning and Management” (Mean Course Effectiveness: 89.5/100)
- ◆ 2025 **Guest Lecturer** of HKU GEOG 7001 “Survey and c in Transport Studies”
- ◆ 2025 **Instructor** of HKU workshop: Introduction to Mobile Mapping (Teaching Performance Score: 84.4/100)

Skills

English: IELTS 7.0

Cantonese: Basic

Mandarin: Native

Programming Language: Python, C/C++ (National Computer Rank Examination Level 2), and MATLAB

Projects

- ◆ 2025.09-2027.08 **Project Manager** of the STF funded ‘Bus Travel Time Prediction Algorithm for Handling Complex Road Conditions’ (PSRI/110/2501/PR)
 - Developing a predictive algorithm capable of handling complex road conditions and accurately forecasting bus Estimated Time of Arrival (ETA) based on machine learning and big data technologies
- ◆ 2025.01-2026.12 **Principal Investigator** of the Wuhan University Open Fund of State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing ‘Research on Risk Assessment Method of Large-scale Transportation in High-density Urban Environment’ (24I03)
 - Developing an object detection and tracking technology based on visual and point cloud data, as well as an interaction behavior recognition technique, to offer feasible research paradigms.
 - Establishing a multi-dimensional analysis framework to identify the relationship between dynamic environmental and traffic factors and driving behaviors under the impact of long and heavy vehicles in high-density urban environments.
- ◆ 2024.02-2025.03 **Co-Project Manager** of the HyD funded project ‘Provision of Services

for Feasibility Study on Simultaneous Localization and Mapping (SLAM) -based Mobile Mapping Technology for Road Image Capturing in Urban Canyons to the Highways Department' (Hy(S)Q/132/2023)

- Developing the multi-fusion scheme that fuses GNSS/ INS, LiDAR-based SLAM, and visual SLAM to improve vehicle positioning accuracy in urban canyons.
- Developing the trajectory adjustment algorithm based on the ground control points to improve the vehicle positioning accuracy in GNSS-denied and feature-less environments.
- ◆ 2024.07-present Core members of the STF funded 'A Smart Planning Platform for Safe and Efficient MiC Module Transport' (PSRI/69/2306/RA)
 - Modeling traffic characteristics under the impact of the MiC trailers.
 - Developing the microscopic simulation for traffic impact review in critical road segments.
 - Develop the traffic impact analysis function of the smart planning platform.
- ◆ 2020-2024 Core members of the ITF funded 'Automatic body temperature measurement and high-risk areas and crowds identification system' (SST/051/20GP), ITF funded 'A Decision-Support Platform for COVID-19 Pandemic Control' (ITP/041/21LP) and the RGC-CRF funded 'Spatiotemporal prediction and real-time early warning of COVID-19 onset risk' (C5079-21GF)
 - Taking charge of the mapping part of the project.

Work Experience

- ◆ 2022.12-2024.06 ITF funded **postdoctoral fellow** of The Hong Kong Polytechnic University
 - Applying for research grant applications, with **one successfully awarded**.
 - Developing the vehicle mobile mapping system prototype, which aims to integrate high-accuracy GNSS/INS and multi-line LiDAR to build a sub-centimeter level point cloud map.
 - Developing the technology of extracting global principle planes and using these planes to constrain the error divergence of the backpack mobile mapping system in indoor environments.
- ◆ 2018.10-2019.06 Autonomous driving **algorithm engineer** of Wuhan ZhongXiang Technology Co., Ltd.
 - Developing a wheeled robot that can automatically locate and navigate using the SLAM algorithm.
 - Developing the positioning module of an autonomous driving car prototype.
 - Developing the autonomous parking algorithm.

Study Experience

- ◆ 2019.9-2022.10 PhD candidate of The Hong Kong Polytechnic University
 - Developing the backpack, handheld, and robotics mobile mapping system prototype.
 - Developing the extrinsic calibration algorithm between GNSS sensor, IMU and LiDAR with high accuracy and repeatability.
 - Developing a systematic mapping framework for the backpack mobile mapping system in common monotonous environments, improving the accuracy from meter-level (existing state-of-the-art methods) to decimeter-level or even centimeter-level in such environments.
 - Developing a tight coupling mapping method that integrates ESKF, g2o, and the point

cloud alignment with centimeter-level accuracy.

- ◆ 2015.09-2018.04 Master candidate of Nanjing University of Aeronautics and Astronautics
 - Autonomous navigation method for quadrotors based on aerodynamic model. The parameters of the model are identified using a Kalman Filter. The aerodynamic model, INS, and LiDAR are fused by Federated Kalman Filter to provide reliable localization under the GNSS-denied circumstance.
 - The fault diagnosis is based on the residual Chi-squared test.